

Patterns and outcomes of traumatic neck injuries: A population-based observational study

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ABSTRACT

Objectives: We aimed to analyze the pattern and outcome of traumatic neck injury (TNI) in a small population.

Materials and Methods: It is a retrospective analysis of all TNI patients who were admitted to the trauma center between 2008 and 2012. Patients' demographics, details of TNI, associated injuries, hospital course, and mortality were analyzed.

Results: A total of 51 TNI cases were included revealing an overall incidence of 0.61/100,000 population. The mean age was 31 ± 9 years. The most frequent mechanism of injury was motor vehicle crash (29.4%) followed by stab (17.6%), machinery injury (17.6%), fall (9.8%), and assault (7.8%). Larynx, thyroid gland, trachea, jugular veins, and carotid were the commonly injured structures. The majority of cases had Zone II TNI whereas isolated injury was observed in 11 cases. TNI were mainly presented with active bleeding (38%), hypovolemic shock (16%) and respiratory distress (16%). Surgical interventions mainly included simple repair and closure (53%), vein ligation (12%), repair of major arteries (4%), tracheal repair (6%), larynx and hypopharynx repair (4%), and repair of parotid gland (2%). Neck exploration was performed in 88%, and emergency tracheostomy was required in 18% of cases. Overall mortality rate was 11.8%, of which five patients had associated injuries, and one had isolated TNI. **Conclusion:** TNI are not frequent but represent an alarming serious entity in Qatar. Patients with persistent signs of major injuries should undergo early operative interventions. Moreover, the effective injury prevention program should be developed to minimize these preventable injuries in the majority of cases.

Key Words: Complications, management, neck injury, outcome, trauma

INTRODUCTION

Traumatic neck injuries (TNIs) represent 5-10% of all severe trauma cases. Particularly, neck injuries related to motor vehicle crashes (MVCs), suicide, and homicide accounted for 3500 deaths annually.^[1] Both blunt and penetrating neck injuries (PNIs) are associated with poor outcomes. Blunt trauma causes around 5% of all neck injuries

which are mainly associated with MVC-related neck injuries usually occur due to the sudden collision of the neck with the steering wheel or dashboard which might cause tracheal crush at the cricoid ring or esophageal compression against the cervical vertebrae.^[1] A retrospective study from Pakistan reported 15 neck trauma cases of 5 (33.3%) patients were involved in MVCs.^[2]

The authors observed poor outcome in 20% of the cases who had delayed surgical intervention. Furthermore, a prospective study from Bangladesh reported 18% of cut throat cases secondary to accidental injuries.^[3] Moreover, cut throat injuries secondary to MVCs and fall-related injuries can also be caused by a pointed object which leads to penetrating injury to a lesser extent. These injuries occur mainly due to glass or sharp metallic projection inside the vehicle.

In blunt trauma patients, injuries of the laryngotracheal (0.04%) and pharyngoesophageal (0.3%) region are relatively

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infrequent as compared to the aerodigestive injuries caused by penetrating neck trauma (5-15%).^[4-6] Though, injuries to the vertebral column (0.7%) and carotid (0.9%) arteries are rare; they often prove fatal in blunt trauma.^[7,8] Interestingly, PNIs are reported in 5% to 10% of all trauma cases.^[9] PNI are mainly related to gunshot wound (GSW) and stab wound. For instance, the majority of PNI in the United States are caused by firearms (44%), stabbing (40%), and GSW (4%) in urban areas.^[10] Particularly, injuries of the aerodigestive tract (23-30%) are frequently observed in PNI cases and of note, esophageal injuries are associated with poor outcomes.^[11] Furthermore, vascular injuries are responsible for complications in around 40% of PNI cases. Cut throat injuries are usually associated with fatal complications and are potentially life-threatening due to the involvement of vital structures. Usually, the PNI are directly proportional to the crime rate and military conflict in a particular country.^[11] Nevertheless, the overall rate of TNI is difficult to assess and usually remains under-reported. To the best of our knowledge, there is no published literature on the presentation of TNI from our region. Therefore, we highlighted the incidence, etiology, injury patterns, and outcome of neck injuries to understand its burden on the healthcare system in Qatar; which is a small country (1.8 million people) with high population diversity (80% are expatriates).

MATERIALS AND METHODS

It is a retrospective analysis of all TNI patients who were admitted to the Hamad General Hospital (HGH) between January 2008 and December 2012. HGH is a tertiary hospital with Level I trauma center in the state of Qatar. Patient with minor neck injury who did not require hospital admission were excluded from the study. Furthermore, brought in dead cases were not included due to incomplete information. Data were reviewed for demographics, mechanism of injury (MVCs, fall from height, fall of heavy object, stab, machinery, self-inflicted, and assault), injury severity score (ISS), neck injuries (involving skin, vascular, nerve, and thyroid gland), zone of injury (Zone I–III), associated injuries, and interventions. Hospital length of stay, ventilatory days, major complications, and mortality were also recorded. Details regarding the hospital course of all patients were recorded from admission until discharge or death. The primary outcome was the all-cause hospital morbidity and mortality.

In order to assess the severity of the injury, the neck has been divided into three anatomic zones. Zone I extends between the clavicles and the cricoid cartilage. Zone II is superior to Zone I and extends as far as the angle of the mandible. Zone III is the area between the angle of the mandible and the base of the skull.^[12] Diagnosis of TNI was confirmed by computed tomography (CT) for stable patients or operative exploration in the presence of one of the following: Clinical signs of neck injury involving the platysma (for penetrating injury), hard signs of vascular injury (bleeding, hematoma, and shock), and signs of aerodigestive injury.

The management of neck injuries in our hospital is based on the guidelines of the Eastern Association for Surgery of Trauma.^[13] The present study has been approved by the Medical Research Center (IRB No. 13270/13) at Hamad Medical Corporation, Doha, Qatar. Data were presented as proportions, medians, or mean $\pm$ standard deviation as appropriate and the analysis was carried out using the Statistical Package for Social Sciences version 18 (SPSS Inc., Chicago, IL, USA).

RESULTS

In this study, a total of 51 TNI cases were included revealing an overall incidence of 0.61/100,000 population. The mean age was 31 ± 9 years and majority (98%) was males. Ninety-two percent of the cases were expatriates (of them 43% were from India and Nepal), and 8% were nationals [Table 1]. MVC ($n = 15$; 29.4%) was the most frequent mechanism of injury followed by stabbing ($n = 9$; 17.6%), machinery injury ($n = 9$; 17.6%), fall from height ($n = 5$; 9.8%), and assault ($n = 4$; 7.8%) [Figure 1].

The most frequently injured structures of neck includes larynx ($n = 6$; 11.8%), thyroid ($n = 6$; 11.8%), trachea ($n = 5$; 9.8%), internal ($n = 4$; 7.8%) and external ($n = 4$; 7.8%) jugular vein, hypopharynx ($n = 3$; 5.9%), and carotid artery ($n = 1$; 2%) [Figure 2]. The isolated neck injury was observed in 11 (21.6%) cases while the remaining 40 cases had other associated injuries. Moreover, head ($n = 35$; 68.6%), chest ($n = 11$; 21.6%), and upper extremities ($n = 12$; 23.5%) were the frequently injured body regions [Figure 3]. Zone II (78%) was the most commonly injured region of the neck. TNI patients were mainly presented with active bleeding (38%), hypovolemic shock (16%), and respiratory distress (16%).

Common surgical interventions mainly included simple repair and closure (53%), vein ligation (12%), repair of major arteries (4%), tracheal repair (6%), larynx and hypopharynx repair (4%), and repair of parotid gland (2%). Neck exploration was performed in 88.2% cases, 38% were intubated and emergency tracheostomy was needed in 17.6% of cases. Intubation through the throat cut was performed in 1 case only. The median length of hospital stay was 4 (range; 1-59) days and Intensive Care Unit stay was 3 (range; 1-36) days.

Permanent tracheostomy was needed in four patients (of which one had persistent dysphagia), one patient developed wound infection, one had an ugly scar and three required psychiatric consultation and long-term care. Cardiac arrest as a mode of presentation was observed in 2 cases. Ipsilateral brachial plexus injury was found in two patients and one patient developed pharyngocutaneous fistula and died. The overall mortality rate was 11.8% (six patients); 5 (83.3%) had associated injuries, and only one (16.7%) had isolated neck injury. Three patients died due to MVC-related TNI, whereas stabbing, fall, and assault accounted for mortality of the other 3 cases.

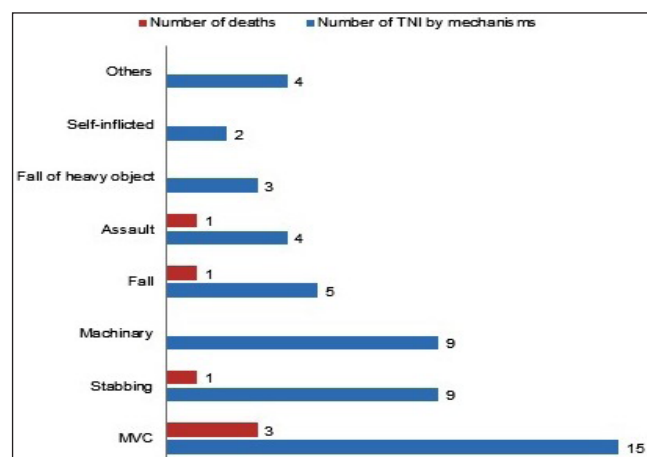


Figure 1: Number of cases with neck injury based on the mechanism of injury and number of deaths

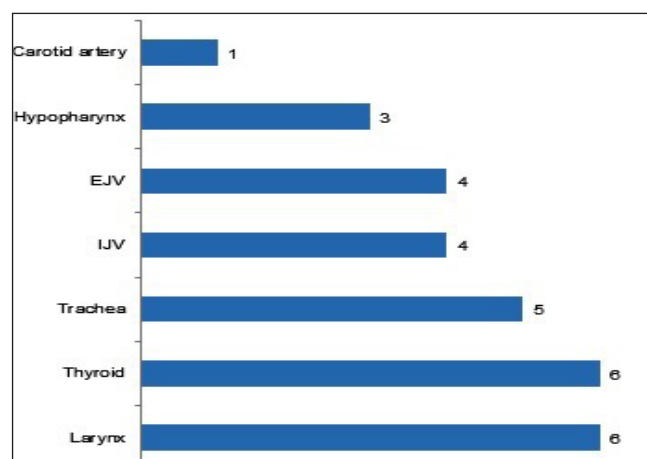


Figure 2: Details of neck injuries

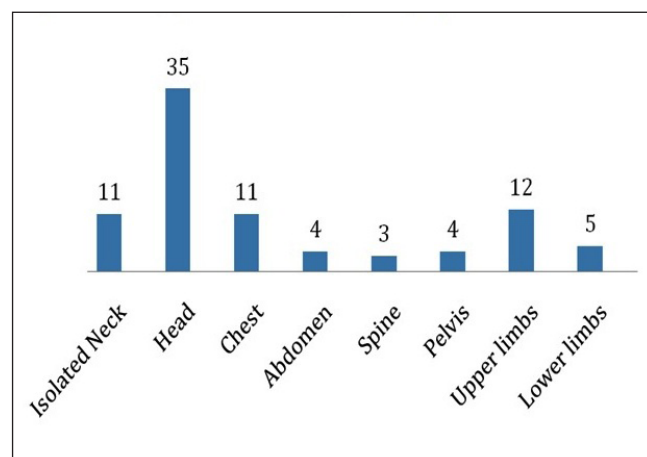


Figure 3: Neck injury and associated injured body regions

DISCUSSION

Despite the fact that TNI can cause potentially life-threatening complications, the exact incidence, management, and outcome of TNI remain underreported. Moreover, there are limited studies

Table 1: Overall analysis of neck injury cases (n = 51)

Variable(s)	Number/proportion
Age (median)	30 (18-56)
Males (%)	50 (98)
Nationality (%)	
Indian	11 (21.6)
Nepalese	11 (21.6)
Qatari	4 (7.8)
Phillipino	3 (5.9)
Bangladeshi	3 (5.9)
Pakistani	1 (2)
Arab non-Qatari	7 (13.7)
Others	11 (21.6)
Neck injured zone (%)	
Zone I	7 (14)
Zone II	39 (78)
Zone III	2 (4)
Multiple zones	2 (4)
BMI (mean±SD)	24.9±3.4
SBP (mean±SD)	129.8±17.4
DBP (mean±SD)	74.9±16.2
Temperature (mean±SD)	36.8±0.6
Respiratory rate (mean±SD)	19.2±3.2
Pulse (mean±SD)	92±18
Lactate (mean±SD)	2.5±1.2
Active bleeding (%)	19 (38)
Shock (%)	8 (16)
Respiratory distress (%)	8 (16)
Neck exploration (%)	45 (88.2)
Treatment (%)	
Simple repair and closure	27 (53)
Ligation of major veins	6 (11.8)
Ligation of major arteries	2 (4)
Repair of trachea	3 (5.9)
laryngeal fracture treated conservatively	2 (3.9)
Repair of larynx and hypopharynx	2 (3.9)
Repair of parotid gland	1 (2)
Repair of thyroid cartilage	1 (2)
Chest tube for hemothorax	1 (2)
Intubation (%)	19 (38)
Tracheostomy (%)	9 (17.6)
GCS (median)	15 (3-15)
ISS (median; range)	5 (1-58)
Alcohol levels (mean±SD)	30.8±14
Hospital LOS (median; range)	4 (1-59)
ICU LOS (median; range)	3 (1-36)
Ventilatory days (median; range)	2 (1-5)
Mortality (%)	6 (11.8)

BMI: BODY MASS INDEX; SBP: SYSTOLIC BLOOD PRESSURE; DBP: DIASTOLIC BLOOD PRESSURE; GCS: GLASGOW COMA SCORE; ISS: INJURY SEVERITY SCORE; LOS: LENGTH OF STAY; ICU: INTENSIVE CARE UNIT; SD: STANDARD DEVIATION

on neck injuries from the rapidly developing countries like Qatar with a diverse population. To the best of our knowledge, this is a unique study from our region that describes the injury pattern, management, and outcome of TNI. A recent study from London demonstrated the overall incidence of PNI to be 4.3/100,000 populations.^[14] Though, the overall incidence of TNI cases was lower in our series, but it is predominantly observed among the physically active young male population. Our findings are

consistent with earlier reports showing a high incidence of TNI among young males [Table 2].^[2,3,11,15-17] In our series, TNI were mainly related to MVC (29%), machinery injury (18%), stabbing (18%), and falls (10%). Similarly, Akhtar and Awan^[2] observed 15 cases of TNI; of which 33% sustained MVCs, 20% had GSW, 7% had machinery injury. However, Aich *et al.*^[3] reported 67 cases; 48 (71.6%) had homicidal injury, 12 (17.91%) sustained injuries due to accidents, and 7 (10.44%) cases were involved in the suicidal attempt. The authors reported political conflict and land dispute to be significantly associated with homicide. Whereas, MVC is the primary cause of accidental cut throat injuries caused either by broken glass or by the insertion of sharp projection of the vehicle after distortion.

In our series, a higher proportion of TNI cases (78.4%) had associated injuries while 21.6% cases had isolated neck trauma. Moreover, laryngotracheal and hypopharynx injuries were observed in 21.6% and 6% cases in our series which corroborate with an earlier study demonstrated similar incidence of laryngotracheal (24.9%) and pharyngeal (8.2%) injuries.^[11] Contrarily, a prospective study of 223 PNI cases mainly observed vascular, spinal cord, aerodigestive tracts, and nerves injuries.^[10] In our study, head is the most commonly associated injured body regions followed by upper extremities and chest. However, Mahmoodie *et al.*^[11] reported chest injuries (65%) to be the most frequently associated injured body region among TNI patients. Due to the anatomical position, the vast majority of studies identified neck injuries within Zone II (50-80%).^[10,11,18,19] Consistently, the majority of TNI patients in our study sustained injuries in Zone II (78%). Moreover, TNI patients were mainly presented with active bleeding, hypovolemic shock, and respiratory distress in our study. Atkins *et al.*^[6] also reported similar rates of active bleeding and hemorrhagic shock.

Though, the incidence of neck injury is infrequent, proper management of TNI is crucial to avoid major complications. Operative management is indicated in patients with persistent signs of major injury such as active hemorrhage, hoarseness, stridor, respiratory distress. In our study, the majority of the cases underwent neck exploration followed by intubation and emergency tracheostomy. An earlier study reported similar rates of tracheostomy but showed a higher rate of intubation.^[11] Particularly, the appropriate selection of diagnostic approach for TNI cases remains controversial. Since, long-time management of TNI patients with platysma injury advocates mandate surgery. However, this concept has been changed in many trauma centers to support selective nonoperative management.^[12]

Moreover, systematic clinical examination remains the cornerstone based on a stepwise protocol in which the selected investigations will be confirmatory. Hence, selective observation with findings of physical examination and symptoms is getting more attention.^[20,21] Consistently, many studies have also proposed selective observation to rule-out major vascular or esophageal injuries.^[22,23] A recent review on the management of PNIs also supports selective nonoperative management and advocated the use of CT angiography for the detection of potential vascular injuries.^[24]

In our study, TNI patients had a lower injury severity with median ISS of 5 (range; 1-58) which is also reflected by the shorter median hospital stay 4 (range; 1-59) days. Consistently, Aich *et al.*^[3] reported lower injury severity in the majority (81%) of the TNI patients who were discharged from the hospital within 2 weeks of admission. In our series, the overall mortality rate was 11.8% and of the six patients died, five had associated injuries and one sustained isolated neck injury. Similar, rates of mortality have been observed by different studies from Nigeria (10%) and Bangladesh (9%).^[3,15,16] In contrast, Mahmoodie *et al.*^[11] reported

Table 2: Review of neck injury cases

Authors	Country	Duration (years)	Study design	Number of cases	Average age	Male (%)	Mechanism of injury %	Outcome
Akhtar and Awan ^[2]	Pakistan	5	Retrospective	15	33	80	Accidental=40 Homicide=33.3 Suicide=26.7	Poor results in 3 cases who had delayed surgical intervention
Aich <i>et al.</i> ^[3]	Bangladesh	1.5	Prospective	67	29	70	Homicide=71.6 Accident=17.9 Suicide=10.4	Mortality=8.9%
Mahmoodie <i>et al.</i> ^[11]	Iran	10	Retrospective	192	25	96.4	Stab=86	The majority of patients were young men Mortality=1.5%
Iseh and Obembe ^[25]	Nigeria	10	Prospective	19	21	100	Suicide=52.6 Homicide=26.3 Accident=10.5	Mortality=10.5%
Onotai and Ibekwe ^[26]	Nigeria	15	Retrospective	24	26-45	100	Suicide=60	58% had complications; Mortality=8.3%
Ozdemir <i>et al.</i> ^[27]	Turkey	10	Retrospective	15	28	60	Honor homicide was predominant	—
Present study	Qatar	5	Retrospective	51	31	98	Accident=30.6 Occupational injury=18.4 Suicide=16.3 Stab=10.2 Fall=10.2	Mortality=11.8%

a lower mortality rate (1.5%) which was mainly associated with other injuries, rather than that of PNI.

One of the major limitations of this study includes its retrospective design which attributes to some missing information regarding the alcohol abuse and intervention details. Furthermore, the smaller sample size is another limitation which is described by the lower incidence of neck injuries in our population. Hence, the expected variation of our findings with current literature might be due to small numbers, rather than to real differences. Moreover, machinery, self-inflicted injury, and assault can be caused either by a blunt or penetrating trauma, and lack of this information renders us to classify our cases into blunt and penetrating injury mechanisms. Furthermore, this study lacks clinical follow-up to look for long-term outcomes. Furthermore, prehospital deaths were not included in this report.

SUMMARY AND CONCLUSION

Though TNI is not frequent in Qatar, it represents an alarming entity due to its fatal complications if remains untreated. The majority of TNI cases are young males, mainly injured by MVCs, occupational injuries, and stabbing. Patients with persistent signs of major injuries as evidence by active hemorrhage and respiratory distress should be preferably managed by operative intervention. Moreover, early neck exploration should be done in patients with deeper neck injuries to avoid severe complications and worst outcomes. Interestingly, the overall complication rate was low in our patients, and the observed morbidity and mortality are mainly attributed to the associated injuries rather than isolated TNI. Hence, early interventions and injury prevention based on major risk factors are important for preventing these injuries. Moreover, the findings of the present study could be the basis for the motivation of public health authorities to initiate awareness programs focusing on socio-cultural, mental health, and psychiatric consultation.

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